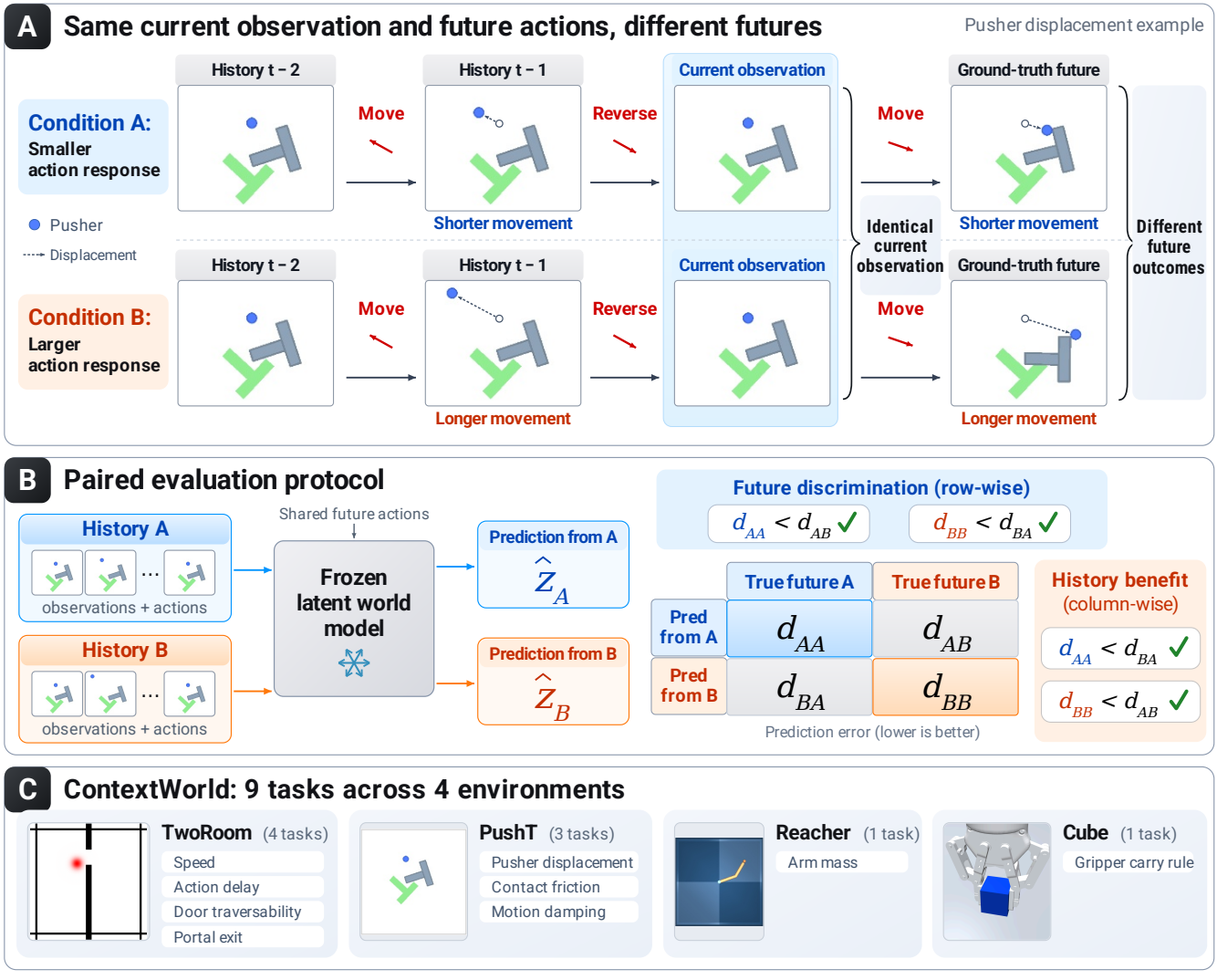

A
Same current observation and future actions, different futures
Pusher displacement example
Condition A:
Smaller action response
Pusher
Displacement
Condition B:
Larger action response
History t − 2
Move
History t − 1
Shorter movement
Reverse
Current observation
Move
Identical current observation
Ground-truth future
Shorter movement
Different future outcomes
History t − 2
Move
History t − 1
Longer movement
Reverse
Current observation
Move
Ground-truth future
Longer movement
B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Shared future actions
Frozen latent world model
Prediction from A
$\hat{Z}_A$
Prediction from B
$\hat{Z}_B$
Future discrimination (row-wise)
$d_{AA} < d_{AB}$ ✓
$d_{BB} < d_{BA}$ ✓
True future A
True future B
Pred from A
Pred from B
d_{AA}
d_{AB}
d_{BA}
d_{BB}
Prediction error (lower is better)
History benefit
(column-wise)
$d_{AA} < d_{BA}$ ✓
$d_{BB} < d_{AB}$ ✓
C
ContextWorld: 9 tasks across 4 environments
TwoRoom (4 tasks)
Speed
Action delay
Door traversability
Portal exit
PushT (3 tasks)
Pusher displacement
Contact friction
Motion damping
Reacher (1 task)
Arm mass
Cube (1 task)
Gripper carry rule